

Solmaz Haddady, MSc

Machine Learning Engineer | Computer Vision | Deep Learning |
Multimodal AI

Linz, Austria

+43 6606408565

solmazhaddadi4003@gmail.com

LinkedIn: [linkedin.com/in/solmaz-haddady-8a94801a3](https://www.linkedin.com/in/solmaz-haddady-8a94801a3)

GitHub: <https://github.com/solmazhaddady>



PROFESSIONAL PROFILE

AI Engineer and Medical AI researcher with expertise in deep learning, computer vision, multimodal learning, and large-scale medical image analysis. Experienced in designing end-to-end AI pipelines—from data curation and preprocessing to model development, evaluation, and deployment-oriented research—for real-world clinical applications. Specialized in Vision–Language Models, whole-slide imaging, and foundation models, with hands-on experience collaborating with clinicians to develop trustworthy AI solutions for healthcare. Passionate about translating cutting-edge AI research into scalable, impactful technologies for medicine, healthcare and industry.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer & Medical AI Researcher

Department of Dermatology and Venereology, JKU Linz | 2023 – 2025

- Designed and developed end-to-end deep learning pipelines for large-scale medical image analysis, supporting automated diagnosis and clinical decision-making using real-world dermatopathology data.
- Designed CNN–Transformer architectures for whole-slide image classification and rare pattern recognition, achieving 96.65% classification accuracy on clinically validated datasets.
- Implemented rigorous model evaluation, statistical validation, and benchmarking to ensure reproducible and clinically reliable AI performance.
- Developed multimodal Vision–Language Models integrating whole-slide image representations with large language models for automated clinical report generation.
- Built scalable preprocessing, quality control, and feature extraction pipelines for 3,000 whole-slide images, supporting AI development and public dataset preparation.
- Collaborated closely with dermatologists, pathologists, and clinical researchers to translate medical requirements into AI-driven diagnostic solutions.
- Integrated heterogeneous medical imaging and structured clinical metadata into unified machine learning workflows for multimodal analysis.
- Designed robust data preprocessing, normalization, and quality-control workflows to improve dataset consistency and downstream model performance.
- **Research Contributions**
 - Contribution to preparation and curation of a dermatopathology dataset for public release on The Cancer Imaging Archive (TCIA)

AI Research Assistant – Digital Pathology

Department of Pathology & Molecular Pathology, JKU Linz | 03/2023 –10/2023

- Contributed to the development of AI methods for computer-assisted diagnosis in digital pathology, focusing on tumor regression grading and clinical image analysis.
- Organized and processed clinical imaging datasets while supporting data annotation, validation, and preparation for machine learning workflows.
- Collaborated within an interdisciplinary research team to develop clinically relevant AI solutions under real-world healthcare constraints.

Data Processing & Seismic Operations

OEOC Co., Tehran, Iran | 2014 – 2016

- Applied signal processing and time-frequency analysis to seismic data for gas detection.
- Processed and analyzed large-scale time-series datasets.
- Worked with quantitative methods for signal interpretation and pattern analysis.
- Developed experience working with high-dimensional scientific data.

Tutor for Physics & Mathematics

Self-Employed, Tehran, Iran | 2014 – 2017

- Instructed university and high school students in mathematics and physics.

SELECTED AI PROJECTS

DermPath-LLaMA

Multimodal Vision–Language Model for Automated Report Generation

Python | PyTorch | Transformers | LoRA | LLaMA

- Developed an end-to-end multimodal AI system integrating computer vision and large language models.
- **Repository:** https://github.com/solmazhaddady/DermPath_LLaMA_VLM

NMSC Whole-Slide Imaging Dataset

Large-Scale Imaging_Text Dataset & TCIA Preparation

Python | WSI | DICOM | OpenSlide | Pandas

- Built scalable data preprocessing, quality control, and metadata standardization pipelines for large-scale image datasets.
- **Repository:** <https://github.com/solmazhaddady/NMSC-TCIA-Dataset>

WSI-to-DICOM Converter

Image Format Conversion Pipeline

Python | DICOM | OpenSlide | wsidicomizer

- Developed an automated pipeline for converting proprietary whole-slide image formats (SVS, MRXS) into the DICOM standard.
- Improved interoperability between digital pathology platforms and AI research workflows.
- Designed reusable software components supporting standardized medical imaging research.
- **Repository:** github.com/solmazhaddady/WSI-to-DICOM-Converter

Whole-Slide Image Quality Control Pipeline

Automated Image Quality Assessment Pipeline

Python | Computer Vision | Image Processing

- Developed automated preprocessing and quality-control workflows for large-scale datasets.
- Implemented image validation, metadata verification, and data curation utilities supporting reliable AI training datasets.
- Designed reusable tools for scalable data preparation.

EDUCATION

Master of Science in Artificial Intelligence

Johannes Kepler University (JKU), Linz, Austria | 2022 – 2025

- Specialization: Machine Learning, Deep Learning, Computer Vision, Medical Imaging
- Master's Thesis: Classification and Vision–Language Modeling for Skin Cancer Histopathology (Grade: excellent)
- Presented research at the **9th & 10th ÖGDV Science Days (2024 & 2025), Austria**

Bachelor of Science in Bioinformatics

Johannes Kepler University (JKU), Linz, Austria | 2017 – 2022

- Interdisciplinary studies combining Computer Science and Molecular Biology
- Erasmus+ Exchange Year: University of South Bohemia, Czech Republic.

Master of Science in Geophysics:

Tehran University, Iran | 2015 – 2017

- Focus on numerical modeling, wave physics, and earth sciences.
- Master's Thesis: Gas and low-frequency shadows detection with S-transformer time-frequency method (Grade: excellent)

Bachelor of Science in Physics

Tehran, Iran | 2012 – 2015

- Foundational studies in theoretical physics and mathematics.

Research Activities

- **Poster Presentation – 9th ÖGDV Science Days, Bad Gastein, Austria (2024)**
- **Poster Presentation – 10th ÖGDV Science Days, Werfenweng, Austria (2025)**

TECHNICAL SKILLS

Programming

- Python • R

Machine Learning & AI

- PyTorch • TensorFlow • Scikit-learn • NumPy • OpenCV • Deep Learning • Computer Vision • CNNs • Vision Transformers • Vision–Language Models (VLMs) • Large Language Models (LLMs) • Multimodal Learning • Generative AI

Medical Imaging

- Whole-Slide Imaging (WSI) • DICOM • Digital Pathology • Medical Image Analysis • OpenSlide • Wsidicomizer

Data Engineering

- Data Preprocessing • Feature Extraction • Data Curation • Quality Control • Statistical Validation • Clinical Metadata Integration • Large-scale Data Pipelines

Development Tools

- Git • Linux • Docker • CUDA • Jupyter Notebook • VS Code • PyCharm

Research & Deployment

- Model Evaluation • Benchmarking • Scientific Computing • Reproducible Research • Clinical AI

SELECTED PUBLICATION & MANUSCRIPTS

Haddady, S., et al. (Manuscript in preparation).

A High-Resolution Dataset of Non-Melanoma Skin Cancer Whole-Slide Images for Machine Learning Research.

Planned submission to *Scientific Data* (Nature Portfolio).

LANGUAGES

- English: Professional Working Proficiency
- German: +B2
- Persian: Native